nelson

UNITED STATES DEPARTMENT OF AGRICULTURE

FOREST SERVICE

STUDY PLAN

Red alder as a control for *Poria weirii* root rot of Douglas-fir

by

Everett Hansen and Earl E. Nelson

Pacific Northwest Forest & Range Experiment Station
Forestry Sciences Laboratory
Corvallis, Oregon

U.S. GOVERNMENT PRINTING OFFICE : 1968 O—298-607

6200-7 (4/68)

STUDY PLAN NO.

RED ALDER AS A CONTROL FOR *PORIA WEIRII* ROOT ROT OF DOUGLAS-FIR

by

Everett Hansen and Earl E. Nelson

A cooperative study with International Paper Company, Longview Fiber Co., Crown Zellerbach Corp., Oregon State University, and the United States Forest Service.

Approved by:

Robert 7. Tarrant
Deputy Director
PNW Experiment Station — 12/18/74 (date)

J.H. 1/27/75
David Mote
DIVISION FOREST MANAGER
F.a.m. 1/24/75
International Paper Co. — 4/27/75 (date)

R A Brandes
Supv. Forest Mgmt. Sciences
Crown Zellerbach Corp. — 2/7/75 (date)

Carl H. Stoltenberg
Oregon State University — 3-4-75 (date)

Reviewed by:

Richard Krebill	USFS	8/21/74
Richard Miller	USFS	9/5/74
Edmond Packee	MacMillan Bloedel Ltd.	7/24/74
John Hammons	International Paper Co.	9/25/74
Gordon Wallis	Canadian For. Service	7/30/74
D. L. Handley	MacMillan Bloedel Ltd.	7/24/74
David Graham	USFS	7/26/74
Dean DeBell	Crown Zellerbach Corp.	7/16/74
Robert Strand	Crown Zellerbach Corp.	8/6/74
William Ferrell	Oregon State Univ.	8/6/74
Larry Weir	Oregon Dept. For.	8/22/74
John Hazard	USFS	11/8/74
James Stewart	USFS	7/19/74
James Trappe	USFS	7/25/74

Background and justification

Immediate replanting with Douglas-fir on a site infested with Poria weirii will result in continued losses in the new stand, but tested management alternatives for these sites are not available. Research indicates that red alder, in addition to not being susceptible to Poria, produces chemical by-products which are inhibitory to the fungus[1/] and preliminary field examinations have indicated less mortality[1/] and slower fungus spread after inoculation (Wallis, personal communication) on pole size and larger Douglas-fir growing in mixture with red alder. In four 12-17 year-old Douglas-fir plantations on the Siuslaw National Forest, however, alder of the same age growing with the fir did not reduce Poria caused mortality.[2/] The practical effectiveness of alder for control of P. weirii root rot must be determined now.

This study will be supplemented by other planned tests of the effectiveness of red alder to reduce P. weirii losses, established with the cooperation of Port Blakely Mill Co. and Mapleton Ranger District, Siuslaw National Forest. Results of these studies should be applicable to a wide range of conditions in the coastal Douglas-fir region.

1/ Trappe, J. M. 1972. Regulation of soil organisms by red alder: a potential biological system for control of Poria weirii. In A. B. Berg ed. Managing Young Forests in the Douglas-fir Region. pp 35-46. Oregon State University, Corvallis, Oregon

2/ Hansen, E. M. 1975. Incidence of Phellinus (Poria) weirii root rot in 10-17 year old Douglas-fir, alder stands. USDA Forest Service. PNW Res. Note (In Press).

Objectives

We will determine in a controlled field situation whether or not 10 or 20 years of red alder on a site will reduce Poria root rot in subsequent generations of Douglas-fir. The principle objective has been divided into more specific questions.

1. Will 10 or 20 years of red alder grown on a harvested Douglas-fir site infested with P. weirii significantly reduce losses from Poria in subsequent Douglas-fir rotations on that site?

2. Is the effect of red alder on Poria survival and infectivity greater than that produced by a non-nitrogen-fixing hardwood (cottonwood) in the same period of time?

To meet these objectives, the following treatments will be randomly applied after harvest of the infested stand:

A. Plant alder, grow 20 years, harvest, and plant fir.

B. Plant cottonwood, grow 20 years, harvest, and plant fir.

C. Plant Douglas-fir.

D. As in A. but grow 10 years.

E. As in B. but grow 10 years.

Other coordinated studies on the site will determine changes in soil nitrogen, phenols, fatty acids and microorganisms as the 3 stand types develop. Additional information on yield of cottonwood and alder will also become available (see separate study plans). The detailed mapping of the site makes it valuable for future epidemiological and survival studies as well.

Study Site

A 30 acre stand of 45-year-old Douglas-fir owned by International Paper Co. has been harvested in preparation for this experiment. The site, part of the Erickson Tract, is in northwest Oregon between Rainier and Vernonia (T6N, R3W, Sec. 17; see attached road log), at about 1800' elevation. The stand harvested was predominately Douglas-fir, with some western hemlock and red alder scattered throughout. It was fully stocked, except for numerous Poria caused openings. The area is high Site II, with north and east facing slopes ranging from level to 45%. Soils are of the Tolke, Hembre, and Klickitat series. The first two are deep, well-drained soils, while the third, found along the ridge top, is a stoney soil. A detailed soil survey is part of the study record.

There were numerous active Poria infection centers scattered through the stand, and two small "holes" apparently caused by Armillaria mellea. The east corner of the area was more open than the rest, with many scattered trees dead and dying from Poria weirii. This part of the stand seemed to be generally infested with the fungus. The study area was systematically photographed before logging. Thirty-five mm color slides are part of the study record.

Soil samples were collected before logging and analyzed for nitrogen, phenolic compounds, and microorganism populations. Poria weirii was isolated from 84 infected stumps and the clonal relationships of the infection centers determined. These results are in the study record, and representative cultures are in the PNW culture collection.

Plot establishment

The area was logged fall of 1973, using rubber-tired skidders. There was very little soil disturbance, except in plots A_1 and D_{IV} (see map). Slash was broadcast burned spring of 1974, and will be planted winter 1974-75, after a deer fence is completed. Plots will be planted with the appropriate species on an 8' x 8' spacing. This spacing was chosen to maximize chances for roots to contact the old inoculum at an early age, while leaving room to grow at least 10 years without thinning.

Each plot contains 5, 1/20th acre (27' radius) data collection subplots, located to include the maximum number of Poria infected stumps while maintaining a 16' buffer from the plot boundary. Plot corners are marked with 1 1/2" steel pipes, tagged to show their position relative to the 2-chain agrid. Subplot centers are marked with steel fence posts and labeled with treatment number, intensity class, and subplot number. Subplot locations are recorded in the study record.

Treatment assignments

The five treatments are to be replicated five times on the study area; 25 plots have been laid out. These were divided into five groups based on location (the 5 plots in the east corner) or amount of P. weirii inoculum. The 5 east plots were segregated because of the apparent difference in P. weirii distribution in this part of the study area. The 5 treatments were randomly assigned to these 5 plots. The remaining 20 plots were ranked by the number of Poria infected stumps lying within the subplots and divided into 4 intensity classes of 5 plots each. A set of treatments was assigned randomly to each intensity class.

Planting stock

To be supplied by International Paper Co.

alder (Alnus rubra) - 1 year tubelings, local seed source.

black cottonwood (Populus trichocarpa) -

Douglas-fir (Pseudotsuga menziesii) - 1 year tubelings, local seed source.

Plot maintenance

Plantations will be well cared for during the 1st 2 years to ensure full stocking in and near subplots.

Conifers will be weeded from the hardwood plots and alder from the cottonwood and fir plots, as needed.

Treatments A, B, D, and E (alder and cottonwood) will be harvested at 10 and 20 years, as appropriate, unless subsequent information suggests other times are more useful. These plots must be replanted to Douglas-fir after harvest, as in Treatment C.

Data collection and analysis

Success of alder and cottonwood treatments will be measured by comparing disease development in Douglas-fir planted after 10 or 20 years of hardwoods (A, B, D or E); and in timber volume produced on these areas in time. Douglas-fir mortality will be recorded biannually by plot, date, diameter, and cause. Harvested basal area for each subplot will be recorded at time of harvest. Final harvest of Douglas-fir stands will be at age 50. Data from the 20 plots in the western portion of the tract will be subjected to analysis of variance to determine treatment effects.

The 5 plots in the eastern portion of the tract differ in damage patterns and will be looked at to see how well they fit in with the data from the west.

Source	d.f.
Blocks	3
Treatments	4
B x T (error)	12
Subplots	80
Total	99

Job responsibilities

Fencing	1974	IP Co.
Grow planting stock	1973-74	IP Co.
Plot layout	summer 1974	PNW
Planting	winter 1974	crews: IP Co. supervision: joint
Weeding	1975-77	IP Co.
Replanting	1976	exam: PNW planting: IP Co.
Data collection	1976 on	PNW
Publication	?	joint

Cost of Study (PNW)

Preliminary ground work	$7000
Plot layout	$1500
Planting supervision	$ 600
Stand exams and data collection (over 40 years)	$8000
Data analysis and publication	$3000

Motor Log to IP-USFS Poria Study Area - Erickson Tract. From Longview, Washington, cross Columbia River on Longview-Rainier Bridge.

0.0 Turn right onto US 30, toward Astoria, Clatskanie

1.5 Turn left at top of hill, Larson Rd. Hudson Park

1.8 Cemetary on left

2.1 Turn right at stop sign

2.6 Keep straight, do not turn left

3.0 Turn left, Apiary, Vernonia Junction

5.9 Keep straight, do not turn left

8.7 Apiary School

9.2 Bear 45° left, follow main road

11.2 Keep straight. 3 gravel roads on right, middle road down to creek and up, joins other road into plot.

11.8 Turn right, across bridge, follow gravel road

12.2 Keep straight, road from other bridge joins

13.1 Road crosses powerline

13.3 Gravel pile on right

13.4 Turn right

14.6 Rock pit right, plot starts on left

15.2 End of road, end of plot, on uphill side